

Spot the Function

E L A + M A T H · W O R D S T U D Y

Mr. H Math · Day 20 · Worksheet 2

WRITE YOUR NAME — one letter per box, straight and neat

Date: _____

Score: ____ / 10

Every ratio table has **two patterns** hiding in it. Reading **down** a column, the numbers grow by adding — that pattern is easy to see, and it is the one that fools people. Reading **across** a row, one number turns into the other by the **same rule every time** — that is the **function**, and it is the one that actually answers questions. Today you learn to spot the across-rule, even when the table is trying to hide it.

DO NOW · SILENT START

3 min

A music channel plays **3 pop songs for every 2 hip-hop songs**.

Pop Songs, x	Hip-Hop Songs, y
3	2
6	4
9	6
12	?

What goes in the ? box: _____ What is being **added** down each column? x: _____ y: _____

TURN & TALK

3 min

Your partner says: "Easy — to go from x to y , just subtract 1." That works for the first row ($3 - 1 = 2$). Test it on **every** row. Does it still work at $6 \rightarrow 4$? At 12?

Frames: "The rule breaks at the row _____ because _____." "A real function works for **every** row, not just _____."

Big idea: one row can fool you. The rule has to pass **every** row.

Section A · Guess the Word

Fill in the blanks to find the name for the rule that turns every input into its output.

F U [] C T I O []

The SAME missing letter goes in both dashed boxes. Write it here: _____

1 A _____ is a rule that turns **every** input (x) into exactly one output (y). In a ratio table, it runs **across** each row: **$y = x \times$ (the same number every time)**.

"Add 2 each time" describes a column. "Multiply by 7" describes the function. Only one of those answers questions like "how many for 63?"

Section B · Down vs. Across

Study the worked example. Then find the across-rule (the function) for three plain tables.

WORKED EXAMPLE · SUMMER CAMP

Camp rule: **7 campers for every 1 adult**.

Adults, x	Campers, y
1	7
2	14
3	21
9	63

! DOWN (pattern)

x: +1, +1 ... then a jump
y: +7, +7 ... then a jump
Breaks when rows skip.

→ ACROSS (function)

$1 \times 7 = 7$ · $2 \times 7 = 14$ · $3 \times 7 = 21$
 $9 \times 7 = 63$ ✓ every row
 $y = x \times 7$

"For every adult there are 7 campers, so **campers = adults $\times$ 7.**" ← full sentence, both quantities named.

2 Test tubes

4 tubes for every 1 student

Students	Tubes
1	4
2	8
3	12
6	?

tubes = students $\times$

? = _____

3 Tacos

6 tacos for every 3 guests

Guests	Tacos
3	6
6	12
9	18
15	?

tacos = guests $\times$ _____

? = _____

4 T-shirts

4 medium for every 2 small

Small	Medium
2	4
4	8
6	12
12	?

medium = small $\times$ _____

? = _____

The 3-step check (use it on every table):

1. Pick the first row. Ask: *what do I do to x to get y ?*
2. Test that same rule on **every** other row.
3. If it passes every row, it is the function. If even one row fails, keep looking.

Section C · The Tricky Ones

Each table below is trying to fool you. Read the whole story first — the story tells you the rule. Then run the 3-step check.

5 The Pizza Party Jump

TRICK: ROWS SKIP

Pizzas, x	Slices, y
1	8
2	16
3	24
10	?

The story: The 6th grade is ordering pizza for a class party. Every pizza from the shop is cut into **8 slices** — no matter how many pizzas you order. The first three rows show 1, 2, and 3 pizzas. Then the order jumps straight to **10 pizzas** for the whole grade.

The trap: Reading down, y goes 8, 16, 24 ... so it's tempting to write **32** (just +8 again). But the x column jumped from 3 to 10, not 3 to 4. The down-pattern only works when the rows go up by one.

Across-rule: slices = pizzas $\times$ _____ ? = _____ Why is 32 wrong? _____

6 The Messy Camp Sign-Up Sheet

TRICK: ROWS OUT OF ORDER

Adults, x	Campers, y
5	35
2	14
9	?
4	28

The story: Camp still keeps **7 campers for every 1 adult**. But this week's sign-up sheet was filled in by different cabins as they arrived, so the rows are in no order at all: a 5-adult cabin, then a 2-adult cabin, then 9, then 4.

The trap: Reading down, the numbers go 5, 2, 9, 4 — there is no down-pattern to follow at all. Students who only look down get stuck. The across-rule doesn't care what order the rows are in.

Test rows 1, 2, 4: $5 \times$ _____ = $35 \cdot 2 \times$ _____ = $14 \cdot 4 \times$ _____ = 28 Rule: $y = x \times$ _____ ? = _____

7 The Donut Shop Order

TRICK: THE RULE SHRINKS

Donuts, x	Boxes, y
60	5
120	10
180	15
300	?

The story: A donut shop packs every order in boxes that hold exactly **one dozen** — 12 donuts, every box, every time. A school orders **300 donuts** for a big event. The table starts with donuts and asks for boxes — so the output is **smaller** than the input.

The trap: Students look for "x something" and can't find a whole number that turns 60 into 5. The rule goes the other way: ask how many donuts fit in **1 box**, then **divide** the donuts by that number.

Rule: boxes = donuts $\div$ _____ Test 120 and 180: _____ ? = _____ boxes

8 Back to the Music Channel

TRICK: NOT A WHOLE-NUMBER RULE

Pop Songs, x	Hip-Hop Songs, y
3	2
6	4
9	6
30	?

The story: The channel from the Do Now plays **3 pop songs for every 2 hip-hop songs** — all day long. By the end of the day it has played **30 pop songs**. y is smaller than x, but not by dividing by a whole number and not by "subtract 1."

The trap: "Subtract 1" works for $3 \rightarrow 2$ only. Instead, think in groups: how many **groups of 3** pop songs? Each group comes with **2** hip-hop songs. So the rule is two steps: **$\div 3$, then $\times 2$** .

Groups of 3 in 30: _____ Hip-hop songs: _____ $\times 2 =$ _____ Check $9 \rightarrow 6$ with the same rule: _____

9 Brother and Sister Birthdays

TRICK: THE IMPOSTER

Andre's Age, x	Kiara's Age, y
1	4
2	5
3	6
10	?

The story: Kiara is exactly **3 years older** than her brother Andre. Every birthday, they both get one year older. When Andre was 1, Kiara was 4. Andre just turned **10**.

The trap: This looks like a ratio table — but it isn't one. $1 \times 4 = 4$, so someone guesses " $\times 4$ " ... then Andre at 10 would make Kiara **40**! The story makes it plain: she's always 3 years older, never 4 times older. Test the " $\times 4$ " idea on the **second** row before you trust it.

Rule: $y = x$ _____ ? = _____ Is this a ratio table? (circle) **YES / NO** because _____

10 Say it like a mathematician. Pick ONE tricky table above. In a full sentence, name both quantities, the across-rule, and the trap it set for you.

Frame: "In the _____ table, _____ = _____ $\times$ _____ for every row. The trap was _____."

REFINE · SUMMARY — Down a column is a **pattern**; across a row is the **function**. Patterns break when rows skip, shuffle, or shrink — the function never does, because the same rule has to pass **every** row. And if the across-rule is **adding** instead of multiplying, it's an imposter: not a ratio table at all. Next: equivalent ratios are just rows that share one function.

☆ **Don't just look down — look across!** ☆

E L A + M A T H

Word Study Meets Number Sense · Day 20 · Spot the Function · Prepares students for Explore Equivalent Ratios (6.RP.A.3a)
Mr. H Math · Day 20 CCSS